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Study of Fatigue among the Educational Stakeholders in South Indian Institutions

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Abstract

Fatiguecanbeunderstoodasthestressdevelopedduetorepetitiveactivitywhichprovestohaveanadverseimplicationonhuman performance. In this research, we strive to determine the factors that contribute to this issue. The research was conducted through interviews and surveys of students and teachers in the same college. The survey gave many reasons for fatigue as the main issue such as travel distance, duration of lectures. Sustainable solutions have been put forth and rigorous analysis has been done which is also presented in thiswork..

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Keywords: fatigue; distance; duration of lectures;

1. Introduction

Being an integral part of the educational system, one could deeply look through the process withheld by the entire educational administration and its stakeholders. The process here focuses on the methodology of knowledge sharing. Everysteptakenbyhumansisasteptowardsperfection;however,wesometimesfailtorealizethatreachingperfection is an iterative process. Our current education system has a lot to offer. Just by tweaking the process to a practical extent, we can get a step closer to perfection. The eagerness to find what stopped us from reaching that level of perfectionpavedthewayforthisresearchwork.Theattempttofindthefactorsaffectingthewholeprocessof

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knowledgedeliveryleadsusto“Fatigue”.Fatigueplaysoneofthemajorrolesinthecurrentscenarioandseemstobe the route cause for many other problems which were prevalent among the students as well as theteachers.

Several factors can bring in a fatigued state among the stakeholders of the educational system. The two main aspects could be tiredness and boredom. Tiredness arises due to multiple factors but boredom is due to repetitive actions. When people become bored they establish less effective short-term memories in the hippocampus and are less likely to consolidate their daily lessons into strong, more permanent memories while they sleep. Being able to know the factorsoffatigueitisalsoimportanttoseehowthesefactorsaffectthedaytodayobjectiveofthestudentsaswellas teachers.

The main objective of the professor is to deliver the content to the students while that of the student is to absorb the contentbeingdelivered.Whenfatiguecomesintothepicture,theattemptstoteachandtolearnarehindered.Fatigue has an impact on the ability of the student to concentrate and grasp the concepts, feels demotivated and in turn has a declining performance. The professors face difficulty in teaching for long hours, especially when he/she is overburdened with administrativework.

ThispaperaimsatfindingsomeofthekeyfactorsofFatigueamongthestakeholdersandanalyzetheeffectsoffatigue. Two major problems were focussed and surveys were conducted to find suitable solutions. The solutions have been effectively discussed and tested to have reducedfatigue.

2. ResearchObjective

1) To find the fatigue factors affecting the performance of students in a South IndianInstitution.

2) To find the fatigue factors affecting the process of teaching in a South IndianInstitution.

3) To select two major factors and find suitablesolutions.

3. ResearchMethodology

The research was conducted in one of the reputed Institutions of the 3rd largest city in the state of Tamil Nadu, Coimbatore.TheUniversityiswellknownforitsengineeringpedagogy.ThisInstitutehasoneofthefinestprofessors andequallyhard-workingstudents.Initially,westartedwithpersonalinterviewswiththestudentsaswellasteachers to find the fatigue factors and those factors were segregated into three groups which are physical, emotional, curriculum and these factors were given suitable heading and was noted that how many people had mentioned the same factor and by which we finalized two major factors and found the solution by taking multiple surveys.The surveys were based on finding the ideal class timing and classes with a break/ without a break.these surveys were taken based onexperience

4. DesignSamples

The research was mainly targeted to the 1st year and 2nd year students of the institute. The sole reason being that the freshers are exposed to a different atmosphere altogether. They are expected to live their life on their own with mere guidance from their parents. Every decision has to be made by his/her will which they are not used to before. This sudden change in their living habits makes them experience every possible factor of Fatigue as they move further.

To find the fatigue factors among students, Personal interviews were taken among hostellers for 10 minutes each. About 25 students were interviewed. By following this we were able to find the different factors that they felt could

contribute to their academic performance because of fatigue. Every single opinion was taken into account and was broadlyclassifiedintothreemaincategories(Physical,EmotionalandPsychologicalfactors).Eachpointwasclubbed into one main heading under the classification. Counts were taken on how many people mentioned the same point. 2 major problems were identified and were thought of as seeking a solution. Most practical and applicable solutions were put forward and the best were chosen foranalysis.

Theanalysiswasmadeintermsofthesurveywerethetargetgroupsweregiven Googleformsandwereaskedtorate thefatiguelevelbasedontheirexperience.Thereviewsandcommentswerecarefullyanalyzedandconclusionswere drawn. The survey for finding the ideal class timings for students was taken at a particular time where they had their classes staring at the expected timings of the day. This was to ensure that they filled the survey based on their momentary experience and not based on theirmemories.

Theresearchalsoextendstoprofessorsofvaryingagegroups.Thiswastotakeintoaccountthechangingmindsetof professors with different levels of experience. Personal interviews were conducted with professors for 15 minutes to clearly understand their perspective. We could obtain a lot of insights into how the effects of fatigue took a different roleintheprofessor'sroutine.Theprofessorswereinterviewedatdifferenttimingssuchasthestartoftheday,break and at the end of the day to have as much randomization as possible. It also proved effective as the professors could put forward different opinions due to different levels of fatigue in duecourse.

5. Insights

Thedataobtainedwerethoroughlyanalysedwiththehelpofpiechartswhicharepresentedbelow.Theresultsofthe personal interview confirmed that the below factors are the reasons for fatigue among collegestudents:

5.1. Physicalfactors:-

1) Timings of the classes and duration (like early morning, afternoon, late eveningetc.)
2) Distance from college tohostel
3) Assignments and projects (paperwork)
4) Attendancepercentage
5) Deprivedsleep
6) Scheduling ofexams
7) Climate and biologicalfactors
8) Food andMess

5.2. Emotionalfactors:-

1) Monday morningblues
2) Communication and languagebarrier
3) Peerpressure
4) Management and decisionmaking
5) Attention andconcentration
6) Ethics andawareness

7) Lack of recreation andrelief

5.3. Factors related to Curriculum:-

1) Flexibility andUpdates
2) Motive andawareness

Thepersonalinterviewswithstudentsproducedastonishingresponses.Therespondentscameupwithalotoffactors thatcouldnotevenbethoughtof.Everypersonhaddifferentissuesaffectingtheirdaytodayactivities.Thecumulative sum of all the effects could be viewed as the effects of fatigue and had indirect effects on their performance. Their non-verbal clues suggested that they felt highly exhausted on a mere recollection of these factors. They faced difficulties in overcoming these factors in their day to daylife.

However, the professors had a different perspective. All the factors that caused fatigue in students seem to not affect the professors. For them, the activities of students in class and their inattentiveness were the only factors of fatigue. Students who struggle to concentrate in class irked the professors and disrupted their flow. Getting their flow back after an interruption was their major challenge and they lose interest once their emotional balance was disturbed. Student's punctuality also played a major role. They had to repeat the same topic for every student that came late to theclass.Surprisinglytheydidnotfeelfatiguedevenwhentheywereexpectedtoperformadministrativework.They did not even feel peer pressure among their group which was expected to be one of the fatiguefactors.

6) Analytic Session andDiscussions

6.1. Survey to Find the Ideal Class timing:-

Itwasassumedthat0correspondstoNofatigue.Inthis,thestudentsfeltnosymptomsoffatigueandwerecompletely involved in the class anddiscussions.

Similarly, scores 4, 5 and 6 correspond to moderate fatigue. In this grade, the students had recognizable levels of fatigue. They found it difficult to stay focused on the class and got distracted even due to the slightest disturbance.

Scores 7, 8 and 9 correspond to extreme fatigue. In this, the students had no interest in listening to the lecture. The fatigue factors had overpowered their will to concentrate in class. "Students of this class are the reason for the disruption in the flow of the class"- as stated by a professor in a personal interview.

Score 10 corresponds to the worst fatigue. "Students are sitting forcefully just for the sake of attendance"- as stated by a student during the survey.

Initially we conducted survey for the classes commencing at 8:30

Fatigue levels at 8:30 AM class

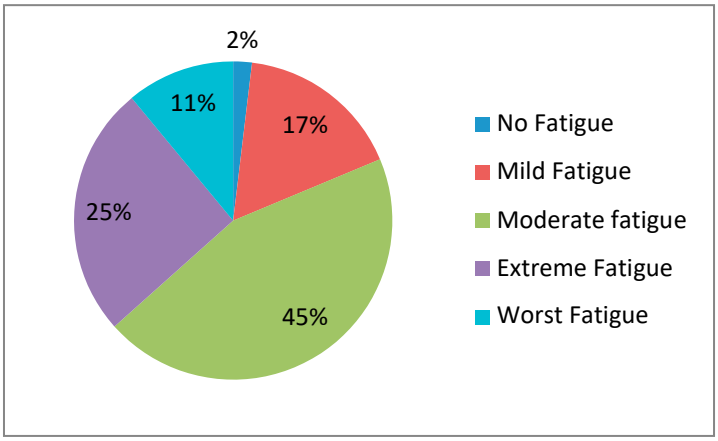

Fig.1

The survey was conducted exactly after the 8:30 class and the students were asked to fill out the survey based on their experience. The responses of 164 students were recorded.

1.9% of respondents felt no fatigue and felt comfortable with classes starting at 8:30 Am.

16.8% respondents felt mild fatigue and were not able to focus for more than 20 minutes of class.

44.7% respondents felt moderate fatigue and were feeling restless.

25.3% respondents felt extreme fatigue and do not want to attend classes for the rest of the day.

11% respondents felt extreme fatigue and were not able to attend the class with focus. They kept talking to their neighbours just waiting for the class to end.

The survey conducted for classes commencing at 10:30 AM

Fatigue levels at 10:30 AM class

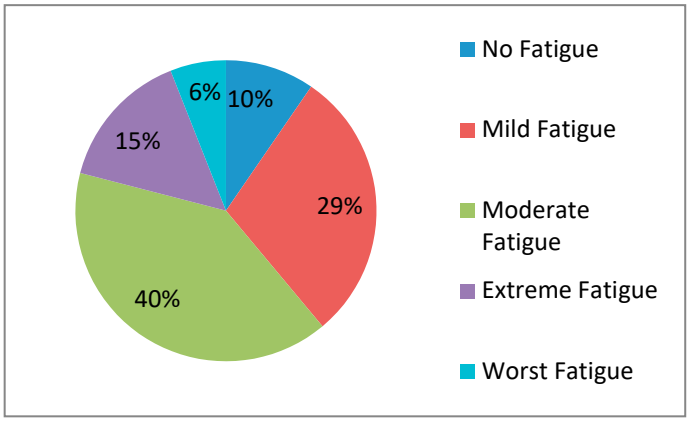

Fig.2

It was ensured that the survey was conducted in the class that started at 10:30AM in that particular day.

9.6% of respondents felt no fatigue and felt comfortable with classes starting at 10:30 Am.

29.40% respondents felt mild fatigue, however, were able to overcome the fatigue factors.

40.20% respondents felt moderate fatigue and were focussed for half of the session.

15% respondents felt extreme fatigue and were completely distracted.

6% respondents felt extreme fatigue and were not able to attend the class with focus. Attendance was their main concern.

The survey conducted for classes commencing at 1:40 PM

Fatigue levels at 1:40 PM class

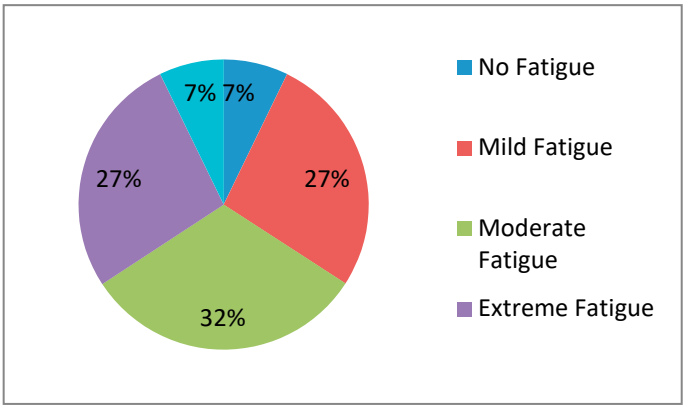

Fig.3

It was ensured that the survey was conducted in the class that started at 1:40 PM in that particular day.

7.2% of respondents categorised themselves to feel no fatigue and were interested throughout the class.

26.9% respondents categorised themselves to feel mild fatigue but had ample amount of focus.

31.6% respondents categorised themselves to feel moderate fatigue.

27% respondents categorised themselves to feel extreme fatigue.

7.2% respondents categorised themselves to feel worse fatigue.

6.2. Analysing the Effects of breaks in-between classes:-

The survey conducted for classes without breaks

Fatigue levels of class withoutbreaks

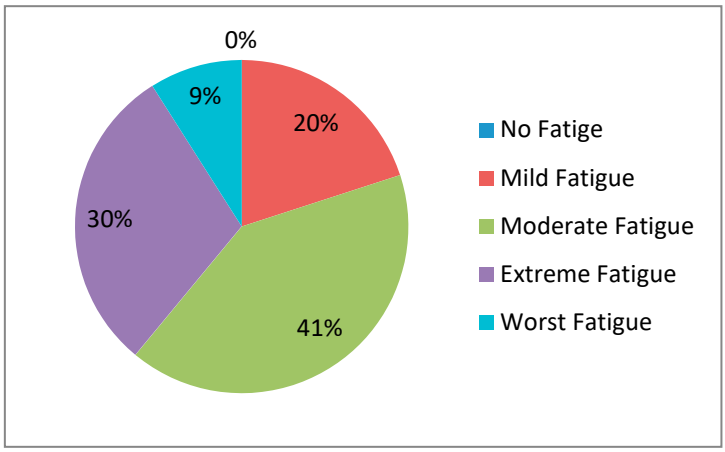

Fig.4

Thedatasuggeststhateveryindividualfeltsomeleveloffatigue.About20%ofrespondentsfeltmildfatigue.41%of the respondents felt moderate fatigue. 30% of the respondents felt extreme fatigue and 9% of the respondents felt worstfatigue.Wecouldconcludethatsomekindofchangewasrequiredinthedurationofclasses.Studentsfelthigh levels of fatigue when they had to attend classes for long durations. Classes extending to long hours and extra hours of classes added to the physical and emotional fatigue. Students also statedthat:
"Alternative methods of teaching and a 2 minute break between new topics would be nice"

The survey conducted for classes with breaks

Fatigue levels for classes with breaks

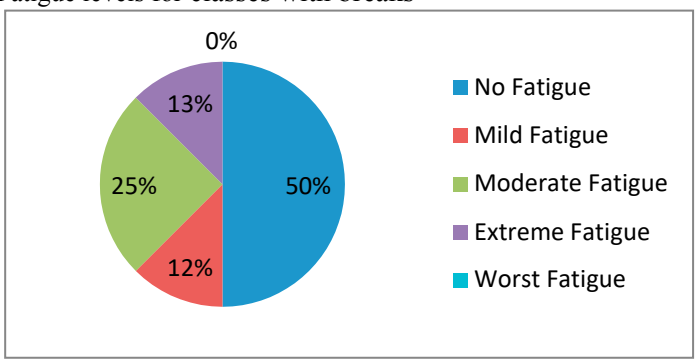

Fig.5

The Data shows that about 50% of the respondents felt no fatigue and were completely satisfied with this method of teaching.About12.5%respondentsfeltmildfatigue.However,intheiropinion,thefatiguefactorswereexternaland were not related to the methodology being tested. 25% of the respondents felt moderate fatigue. They felt tired and exhausted due to the previous classes and could not make through the same. 12.5% respondents felt extreme fatigue. It was astonishing to find out that none of them felt worse fatigue and many liked the approach that was tested. They wanted to follow the same methodology throughout and also stated thebelow:

"Theclassmethodfollowedyesterdaywasveryinteractiveandlookingforwardformoresuchclasseswhichwillhelp us in developingourselves".

The fatigue factors identified for professors through personal interviews are as stated below

Table:1

Questions	Teachers response
What are the few factors that disrupt the process of teaching in view of fatigue ?	• Lossofinterestwhenthestudentsdonotpayanyattention. • The pace of teaching is reduced because of studentsbehaviour. • Deviations or distractions such as mobilephones. • Responsibilities of a commonprofessor.
How these factors could be avoided?	• Keeping jammers in the hall to avoiddistraction • 2-3 mins of break between every newtopic • Reducingworkload • More teacher-studentinteractions
What are curricular aspects to be accounted to avoid the factors of fatigue?	• Classes without breaks are not stressful but more interesting • Update the contentregularly

Conclusion

This research paper highlights fatigue factors affecting performance of students and teachers. Some of the major reasons pointed out by them through surveys and personal interviews as the fatigue factors are the class timings,[4] presence of breaks during class , [2]the distance between the college and the hostel, workload on students and [3]the responsibilities of teachers. From this the class hours and breaks between classes were considered to be the major. From the research findings, solutions were formulated for these two issues[1] The classes should begin only after 10:30 am as it is clear from the findings that the fatigue levels are low at that time. [5]By providing frequent breaks between classes of 2 to 3 minutes fatigue levels are seen to be low.

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